



Lebanese University
Faculty of Engineering

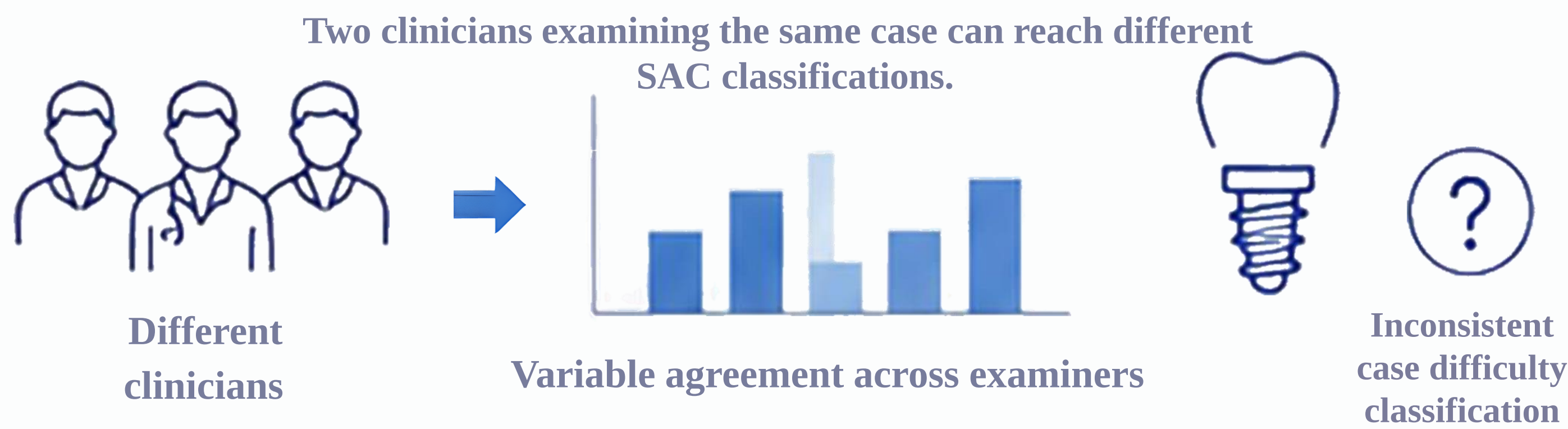
ImplantSAC: AI-Powered Classification of Dental Implant Cases

(Straightforward • Advanced • Complex)

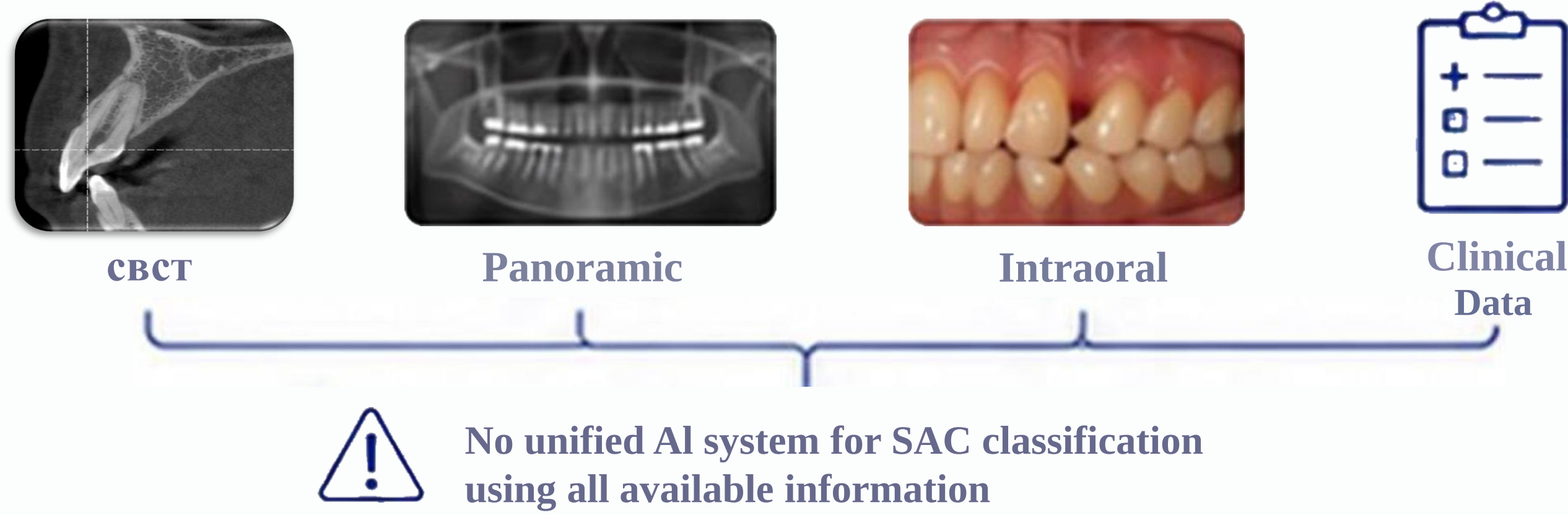


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1 MOTIVATION & PROBLEM



No existing multimodal system



2 OBJECTIVES



Develop and evaluate AI models for classifying dental implant cases into **Straightforward**, **Advanced** and **Complex**.



Leverage four complementary data modalities: **CBCT**, **panoramic**, **intraoral images**, and **clinical data**.



Fuse modality-specific predictions using a **worst-case(max-risk)** rule to ensure patient safety.

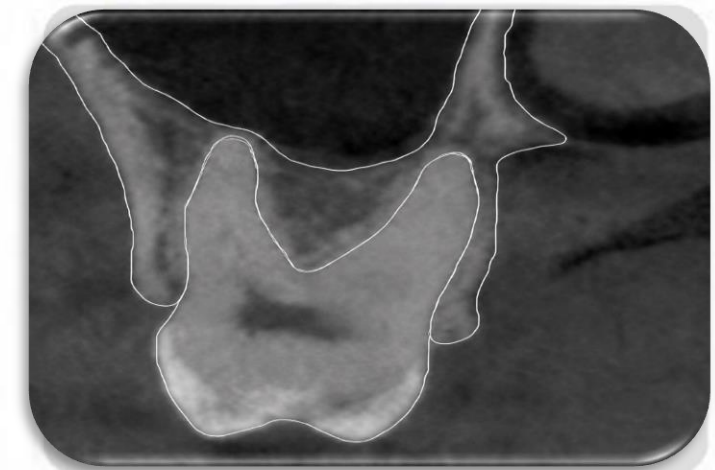


Reduce inter-examiner variability and provide consistent, explainable support for clinical decision-making.

3 METHODOLOGY-IMPLANTSAC MULTIMODAL PIPELINES

1 CBCT PIPELINE

CBCT Sagittal Cut



M1-M5-M7 Router (Per-factor)



5 Bone Factors (Regression/Classification)

Banding



2 PANORAMIC PIPELINE

Panoramic Radiograph



YOLOv8s (Tooth Detection)



Crop + ResNet50 (Lesion Classification)

Outputs per Tooth

- ✓ Lesion Presence
- ✓ Location (Apex/Middle/Crown)
- ✓ Volume Proxy
- ✓ Distance to Sinus/IAC

Banding



3 INTRAORAL PIPELINE

Intraoral Photograph



YOLOv8s (Tooth Detection)

Geometric Proxies (Factors 10 & 11)

ResNet50 (Soft Tissue Quality-F13)

3 Factors

- ✓ Edentulous Span Width
- ✓ Prosthetic Space
- ✓ Soft Tissue Quality

Banding



4 CLINICAL DATA PIPELINE

Clinical & Patient Data



TabNet (Classifier)

SAC Probability Per-factor Attribution



Banding



WORST-CASE(MAX-RISK)RULE (ImplantSAC)

S < A < C

Any Red factor → **Complex**
Any Yellow, no Red → **Advanced**
All Green → **Straightforward**



S A C

Straightforward Advanced Complex

4 KEY RESULTS (BEST PERFORMANCE PER PIPELINE)

1 CBCT PIPELINE (M1-M5-M7 Router)

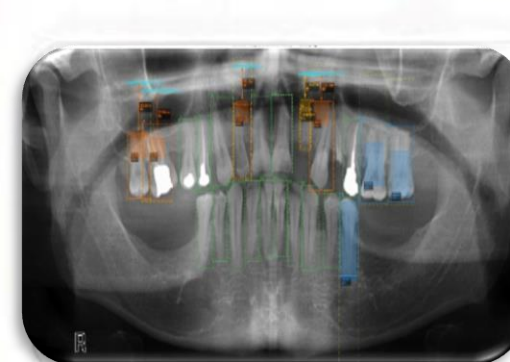
Accuracy(%)

Factor	Strict (Eract)	Tolerance (±margin)
Apical Bone Stability (M7)	92.0	92.0
Buccal Wall Thickness (M7)	67.3	83.1
Horizontal Bone Volume (MS)	83.3	88.1
Vertical Bone Volume (M1)	96.1	96.1
Fenestration (M7)	88.8	-



For each factor, the best-performing model was selected independently.

2 PANORAMIC PIPELINE (YOLOv8s + ResNet50)



Metric	Value(%)
Lesion Presence AUC_ROC	92.73
Lesion Presence F1	82.91
Location Accuracy	86.98



ResNet50 selected as best backbone across all heads: presence, location, and bounding-box regression.

3 INTRAORAL PIPELINE (YOLOv8s + Proxies + ResNet50)

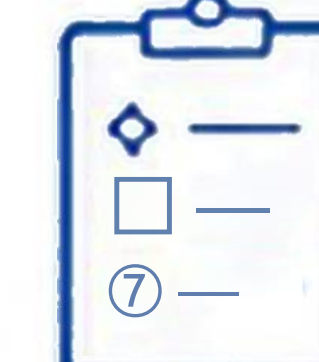


Tooth Detector (YOLOv8s)		Soft Tissue Quality (ResNet50)	
Metric	Value (%)	Metric	Value(%)
mAP@0.5	94.55	AUC-ROC	80.61
Recall	89.67	SEVERE Recall	72.90
		Test Accuracy	72.00



ResNet50 selected over EfficientNet-B4: better SEVERE recall and no residual overfitting.

4 CLINICAL DATA PIPELINE (TabNet)



TabNet: Clinical Factors Pipeline (18 Factors)	
Test Condition	Accuracy(%)
All fields completed	100
Worst-case (only 3 required fields)	95.1



TabNet achieves 95.1% accuracy even in worst-case missingness no Complex case ever misclassified as Straightforward.

Results reported using the worst-case (max-risk) rule where applicable.

5 CONCLUSION & FUTURE WORK

CONCLUSION

- ✓ ImplantSAC leverages four complementary data modalities for robust SAC classification.
- ✓ Worst-case fusion prioritizes patient safety by taking the highest predicted risk.
- ✓ Engineering around the model, curriculum learning, crop-margin correction, calibration, drove performance more than architecture choice alone.
- ✓ ImplantSAC can reduce inter-examiner variability and support consistent clinical decision-making.



FUTURE WORK

- ✓ Expand CBCT dataset beyond 538 images.
- ✓ Replace coordinate prediction with heatmap-based landmark regression.
- ✓ Replace Factor 10 geometric proxy with a direct trained detector.
- ✓ Accept raw 3D CBCT volume as input directly and only.
- ✓ Validate on real clinical patient records.



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